RESEARCH BRIEF · AUTO FINANCE

Toyota Refinance Readiness

What the industry, regulators and academic research say about deciding whether a Toyota owner is well positioned to refinance, and a reference algorithm built from that evidence.

Prepared October 24, 2026 Data through Q2–Q3 2026 Scores: VantageScore 4.0 bands unless noted

Summary

"Well positioned to refinance" is really three separate questions. A credible engine has to answer all three and show its work for each.

1. **Will a lender approve the loan, and at what rate?** This is driven by credit tier, payment history, debt-to-income (DTI), payment-to-income (PTI), loan-to-value (LTV), and the vehicle's age, mileage and title.
2. **Does the new loan beat the old one after all costs, on a like-for-like basis?** Compare term-matched payment streams, net of fees and refunds, discounted to present value. A lower monthly payment alone proves nothing.
3. **Is it a good idea for this borrower?** Guard against term extension that raises total cost, rolling negative equity forward, and replacing a manufacturer-subsidized rate.

Key findings

- In Q2 2026 the average refinanced auto loan went from **10.40% to 7.97%**, saving **$83 a month**. Savings were largest at credit unions ($102), then banks ($65), then finance companies ($38) [1].
- The rate gap between credit tiers is large. For used vehicles it runs from **6.29%** for super prime to **13.93%** for near prime and **19.10%** for subprime [2]. Moving up one tier is the single biggest driver of refinance value.
- Borrowers who keep paying on time become lower risk than their original tier implied. In ABS data, near-prime 72-month loans behave like prime risk after **13–15 months**, and subprime loans after **25–42 months**. Borrowers typically wait too long and forgo **$1,153–$2,327** [3].
- Refinance value is front-loaded. On a 72-month loan at 10%, only **28%** of total interest remains after 36 payments. Timing matters as much as rate.

- Toyota is the strongest mainstream brand for residual value. It took 6 of 26 segments in J.D. Power's 2026 ALG awards [9] and holds 10 of the top 25 spots for 5-year value retention [10]. This makes LTV less often the binding constraint for Toyota owners than for the market overall.
- The Toyota-specific risk is the opposite problem. Many Toyota Financial Services (TFS) loans carry subsidized promotional APRs (for example 3.99–4.99% in 2026 offers [14]) that no refinance lender will beat. The engine must detect these and say "keep your loan."
- 85% of refinancers extend their term. They lower payments by $167 a month but save only $460 in total, against $6,291 for those who shorten the term [4]. The engine should default to term-matched comparisons.

1 Market baseline, October 2026

These figures set the priors for rate estimates. Refresh them quarterly from the primary sources.

Credit tier (VantageScore 4.0)	New APR	Used APR
Super prime (781+)	4.41%	6.29%
Prime (661–780)	6.15%	8.81%
Near prime (601–660)	9.71%	13.93%
Subprime (501–600)	13.52%	19.10%
Deep subprime (300–500)	16.11%	21.62%

Experian, State of the Automotive Finance Market, Q2 2026 [2]. Refinance loans are generally priced off used-vehicle rate sheets.

Indicator	Latest value	Why it matters
Average refinance rate vs. original rate	7.97% vs. 10.40% (Q2 2026)	Typical achievable rate reduction is about 2.4 points [1]
Average new / used loan rate	6.35% / 11.19%	Market-wide reference [1]

Indicator	Latest value	Why it matters
Auto 30-day / 60-day delinquency	2.39% / 0.90%	Rising slightly year over year [1]
Flow into 90+ day auto delinquency	3.00% (Q2 2026) vs. 2.93%	Lenders tighten when this rises [5]
Auto debt outstanding	$1.713 trillion	Size of the refinance pool [5]
Federal funds target	3.75–4.00% (raised Sept. 2026)	Funding costs are rising; the Oct. 28 meeting is priced mostly as a hold [6]
Manheim Used Vehicle Value Index	205.9 (Sept. 2026, −0.6% y/y)	Collateral values flat to soft. Large trucks and SUVs are weakening [7]
Trade-ins with negative equity	29.6%, average $6,884 (Q2 2026)	About 3 in 10 owners start underwater [8]

The rate environment is not falling. A refinance opportunity in late 2026 will usually come from a better credit tier, removing a dealer markup, or leaving a high-rate subprime or finance-company loan. It will rarely come from market rates moving lower.

2 How lenders decide

Refinance underwriting follows the classic credit pillars: character (credit history), capacity (income and debt load), collateral (the car), and the structure of the loan itself. Published thresholds vary by lender. The ranges below are drawn from lender pages and industry guides.

Factor	Typical threshold	Notes and sources
Credit score	580–600 minimum; best rates at 720+	Varies by lender and model version. Lenders may pull FICO Auto Score 8/9/10 (range 250–900), base FICO, or VantageScore [11][12]
Recent payment history	No 30+ day lates on the current auto loan, typically in the last 12 months	Past late payments or accounts not current can trigger a decline [11]
Debt-to-income (DTI)	Below 36% preferred; up to 45–50% accepted by many	Sources range from 35% to 50–60% [11]

Factor	Typical threshold	Notes and sources
Payment-to-income (PTI)	Prime pools 7–9%; subprime caps 15–20%	KBRA reports prime ABS weighted-average PTI of 7–9% since 2023 [13]. S&P links pools with PTI above 10% to rising prime losses [15]
Loan-to-value (LTV)	Up to about 125% at most lenders; best pricing at or below 90–100%	Prime ABS weighted-average LTV is around 90% [13]. Above 100% raises the APR and narrows lender choice [11]
Vehicle age	Under 8–10 model years	"Many lenders will only refinance passenger vehicles under 10 years old" [11]
Mileage	Under 100,000–150,000	100,000 is the common cutoff [11]
Loan balance	Minimum $3,000–$7,500; often $5,000	Small balances produce little savings and lenders decline them [11]
Remaining term	Often at least 24 months remaining	Lenders may require a new term of 24–36 months or longer [16]
Loan seasoning	No fixed rule; 6 months is the common benchmark	Title transfer takes 60–90 days, so the practical minimum is 2–3 months [16]
Title and use	Clean title, personal use	Salvage, flood and rebuilt titles are excluded. Commercial use and heavy modification are restricted [11]

Rates also depend on the vehicle. Navy Federal, for example, classifies a vehicle as "used" by model year and mileage, and prices 73–84 month used refinances about 1.6 points above 61–72 months (7.28% vs. 5.69% as-low-as rates on October 24, 2026) [17]. Term and vehicle age adjustments belong in the rate model, not only in eligibility.

3 Credit tier, migration and seasoning

Scoring models

The score a consumer sees in a free app is often not the score a lender uses. Lenders may use industry-specific FICO Auto Scores (250–900 range), base FICO 8, or VantageScore 3.0/4.0 [12]. Experian publishes its rate-by-tier tables on VantageScore 4.0 [2]. The engine should record the model and version of every score it receives and map it to tiers explicitly. It should not treat all scores as the same number.

Rate shopping does not have to hurt the score

Newer FICO versions count all auto loan inquiries within 45 days as one, older versions use 14 days, and FICO ignores auto inquiries in the 30 days before scoring. VantageScore uses a 14-day window [18]. The product should tell users to submit all hard-pull applications within 14 days. Pre-qualification should use soft inquiries.

On-time payment history lowers real risk

Lautier, Pozdnyakov and Yan (Bentley University and UConn) studied 72–73 month loans from public auto ABS filings. They estimated when the default hazard of one original risk band becomes statistically the same as a better band, conditional on the loan still being current [3].

Original band	→ Near prime	→ Prime	→ Super prime
Deep subprime	50–51 mo	50–58 mo	52–58 mo
Subprime	23 mo	25–42 mo	42–48 mo
Near prime	—	13–15 mo	15–34 mo
Prime	—	—	10 mo

Loan age at which conditional default risk converges. Ranges span the 2017 and 2019 issuance samples (58,118 and 65,802 loans) [3].

The authors estimate borrowers forgo $1,153–$2,327 by delaying a credit-based refinance. They find that near-prime and prime borrowers leave the most money on the table, because they "should refinance quickly, after about only one year" but generally wait longer [3]. For the engine this has two uses:

- **As a trigger.** Months of on-time payments, combined with the original tier, is a strong signal that a borrower is now mispriced. It holds even before the bureau score fully catches up.
- **As a forecast.** For a borrower who is "not yet" ready, the matrix gives a defensible estimate of when they should check again.

This finding concerns risk, and lenders price on the score. The score usually rises with on-time history but on its own schedule. Use the matrix for timing and trigger guidance. Use the actual current score for rate estimates.

4 The vehicle: value, LTV and depreciation

Which value lenders use

Lenders compute LTV against a guidebook value, but they don't all use the same book or the same condition level. Forum and industry reports describe large banks using J.D. Power (NADA) clean trade-in value, while some credit unions use clean retail. Kelley Blue Book and Black Book are also in use [19]. No lender publishes this in a standard way. The engine should therefore:

- compute LTV against a **conservative trade-in or wholesale value** by default, since that is closest to what a lender recovers at auction;
- also show LTV against retail, so the user understands why some lenders will say yes and others no;
- use the numerator lenders use: **10-day payoff amount** plus any fees financed into the new loan, not the statement balance.

Modeling depreciation

The standard parametric form in pricing literature and valuation patents is exponential decay in age, adjusted for mileage and condition [20]. A model-specific decay rate can be calibrated from published 5-year retention:

```
V(t) = MSRP × e^(−δ·t) × m(miles, t) × c(condition) × k(trade)
δ_model = −ln(1 − D5_model) / 5        D5 = 5-year depreciation share
m = 1 − β × (miles − 13,476·t) / 1,000   13,476 mi/yr = FHWA average
```

The constant-rate form understates first-year depreciation and overstates later years. In production, replace it with licensed guidebook values (J.D. Power, Black Book, KBB) by VIN, mileage and options. Keep the curve only for forecasting future LTV. The mileage slope β and the retail-to-trade factor k must be fitted to guidebook data. The calculator uses placeholders (β = 0.35% per 1,000 miles, k = 0.85) that are assumptions, not sourced values. Average annual mileage is from FHWA [21].

Toyota 5-year depreciation

Model	5-yr depreciation	Implied δ / yr	Segment note
Tacoma	19.9%	0.044	#1 truck; #4 overall
Tundra	21.2%	0.048	#2 truck

Model	5-yr depreciation	Implied δ / yr	Segment note
GR Supra	24.0%	0.055	#8 overall
RAV4 / RAV4 Hybrid	25.2%	0.058	#1 small SUV
Corolla Hatchback	25.5%	0.059	
4Runner	25.5%	0.059	#1 midsize SUV
Corolla	27.6%	0.065	
Sienna	28.5%	0.067	#2 hybrid
Corolla Hybrid	30.1%	0.072	
Prius	32.1%	0.077	
Highlander Hybrid	34.3%	0.084	
Highlander	36.0%	0.089	
All vehicles	41.8%	0.108	Trucks 34.2%, hybrids 35.4%, EVs 57.2%

iSeeCars, about 950,000 five-year-old vehicles sold March 2025–February 2026; asking price vs. inflation-adjusted MSRP [10]. Camry, Crown, Sequoia, Grand Highlander and bZ were not reported; use J.D. Power data or a conservative fallback.

J.D. Power's 2026 ALG Residual Value Awards, which forecast future rather than past values, named Toyota the top mass-market brand. Winners were the GR Supra, GR86, Camry, Sequoia, Tundra and Tacoma, with the Tundra winning for a fifth consecutive year [9]. This supports using the Camry and Sequoia at or better than segment averages until guidebook data is wired in.

Negative equity

The CFPB's Auto Finance Data Pilot found that borrowers who roll negative equity into a new loan are **more than twice as likely** to have the account assigned to repossession within two years as borrowers with a positive trade-in. Their payments are 26–27% higher [22]. A refinance with LTV above 100% carries the same structural risk. The engine should treat it as a warning and never present cash-out on an underwater car as a benefit.

5 Toyota-specific factors

Captive financing and subsidized rates

- **Subvented APRs.** Toyota regularly offers below-market APRs through TFS on new vehicles. Examples from 2026 include Camry 4.49% for 48 months, RAV4 4.99% for 48 months, and Tacoma hybrid 3.99% for 48 months [14]. These are at or below super-prime market rates, so a refinance almost never helps. The engine should compare the current APR to the estimated refinance APR first, and return "keep current loan" if the gap is under about 0.5 points.
- **Who borrows from Toyota.** Toyota's own securitized pools skew prime. TAOT 2026-B had a weighted-average FICO of 770, and 83.7% of borrowers were above 700 [23]. The best refinance candidates among Toyota owners are therefore less typical: non-subvented loans arranged by the dealer, borrowers who were near prime at purchase, and loans from third-party finance companies.
- **Dealer markup.** In 2016 the CFPB and DOJ ordered Toyota Motor Credit to cap dealer markup at 1.25 points above the buy rate on loans of 60 months or less, and 1.00 point on longer loans. Before the order, TMCC allowed up to 2.50 points [24]. Industry research finds dealers mark up 78% of loans, by 108 basis points on average [25]. A refinance removes that markup, so a dealer-arranged loan at the borrower's tier rate plus about 1 point is a common, quantifiable opportunity.
- **Payoff mechanics.** TFS retail contracts are generally simple interest, with no prepayment penalty reported. The contract governs, so the engine should ask the user to confirm and fetch a dated payoff quote [26]. For precomputed loans longer than 61 months, federal law requires an interest rebate at least as favorable as the actuarial method, so the Rule of 78s cannot be used [27].
- **Credit performance.** TMCC's net charge-offs were 0.66% of average receivables for the first nine months of fiscal 2026, down from 0.71% [28]. This gives a benchmark for how a Toyota-originated book performs.

Collateral signals

- **Residual strength.** Toyota leads mainstream brands on retained value (section 4). For most Toyota owners, positive equity arrives sooner than for the market as a whole.
- **Hybrids.** Hybrids depreciate less than the market (35.4% vs. 41.8% over 5 years) [10]. Toyota's hybrid battery warranty is 10 years or 150,000 miles for model year 2020 and later, versus 8 years or 100,000 miles before that [29]. A pre-2020 hybrid near or past its battery coverage is a valuation risk.
- **Open recalls.** NHTSA campaign 24V381, expanded by 25V767 and 26V320, covers V35A twin-turbo V6 engines in 2022–2024 Tundra and Lexus LX/GX vehicles. Manufacturing debris can cause main-bearing failure and stalling, and the remedy includes engine inspection or

replacement [30][31]. An open recall rarely blocks a refinance by itself. The engine should still flag it, since appraisals and buyer demand can be affected, and remind the owner to have it repaired.

- **Segment momentum.** In September 2026 Manheim reported weakening wholesale values for large trucks and SUVs while compact cars and EVs held up [7]. For Tundra and Sequoia owners near the LTV limit, a short-term value haircut is prudent.
- **Reliability.** Consumer Reports' 2026 reliability rankings place Toyota at or near the top [29]. This supports lender appetite for higher-mileage Toyotas, but the 100,000-mile cutoff still applies at many lenders.

> **Branding**
>
> Unless the product has an agreement with Toyota, use the Toyota name only to describe the vehicles it covers. Do not use Toyota or TFS logos or styling, and do not imply affiliation or endorsement.

6 Refinance economics

Core formulas

```
r  = APR / 1200                                  monthly rate
A  = P·r / (1 − (1 + r)^−n)                      level payment
B_k = P·(1 + r)^k − A·((1 + r)^k − 1) / r        balance after k payments
Payoff = B + B·(APR/100)/365 × days since last payment   simple interest per diem

PV(A, n, d) = A·(1 − (1 + d/1200)^−n) / (d/1200)
NPV savings = PV(A_old, n_rem, d) − PV(A_new, n_new, d) − upfront costs + refunds
Breakeven months = upfront costs / (A_old − A_new)    term-matched only
LTV = (Payoff + financed fees) / collateral value
PTI = A_new / gross monthly income
DTI = (A_new + housing + other monthly debt) / gross monthly income
```

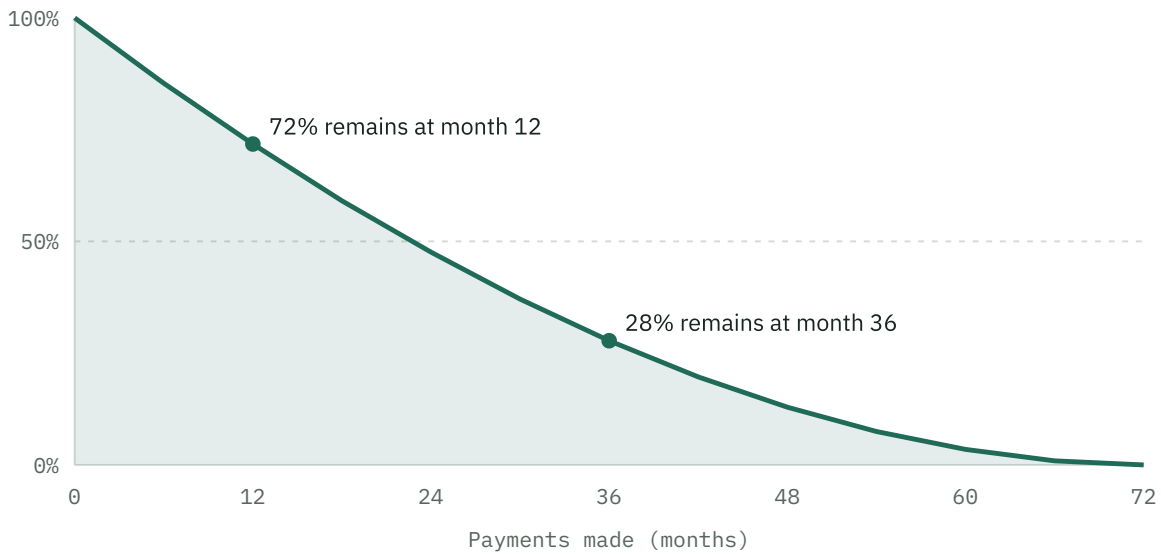
The discount rate `d` should be the borrower's opportunity cost of money, for example a high-yield savings rate, or the rate on their most expensive debt if they carry a credit card balance. Use 5% as a default and let the user change it.

Compare like for like

Always compute the **term-matched** comparison first: new term equal to the months remaining on the old loan. A longer new term lowers the payment and can still raise total cost. Show any term

extension as a separate option, with its total cost next to the term-matched one.

Timing: value is front-loaded



Share of total interest still to be paid on a 72-month loan at 10% APR. A rate cut taken at month 12 applies to almost three times as much interest as the same cut at month 36.

Costs and refunds to include

Item	Typical range	Treatment
Lender origination or application fee	$0 at most direct lenders and credit unions; some charge 1–2%	Upfront cost [32]
Title transfer and lien recording	About $5–$215 by state; re-registration $25–$150 in some states	Upfront cost; keep a state lookup table [32]
Prepayment penalty on old loan	Usually none on simple-interest loans; some contracts 1–2%	Upfront cost; read the contract [32]
State taxes on new notes	For example, Florida's documentary stamp tax of $0.35 per $100 on notes, capped at $2,450	Application to vehicle refinance needs legal review [33]
GAP and service contract refunds	Usually pro rata or short-rate refund on payoff	Refund (lowers net cost), but the borrower may need to buy new GAP if LTV stays high [34]

Item	Typical range	Treatment
Lender refinance bonus	For example, $200 at Navy Federal for loans of $5,000+ open 90+ days	Refund if offered [17]
Interest overlap during payoff	Per diem × days until the old lender receives funds	Upfront cost; usually small

7 What research says about borrower behavior

- **Borrowers optimize the payment, not the rate.** Using 500,000 used auto loans, Argyle, Nadauld and Palmer find demand is more sensitive to loan term than to interest rate. Borrowers bunch at round payment amounts and spend payment savings on bigger loans [35]. A refinance product that leads with "lower your payment" will push users toward longer terms. Lead with total savings.
- **Most people don't shop.** The same authors find search frictions create wide rate dispersion among similar borrowers. The average borrower pays about $488 more in present value by not shopping [36]. Showing offers from multiple lenders creates real value.
- **Longer terms default more.** The CFPB found loans of 72 months or more defaulted at over 8%, versus about 4% for 60 months or less [37]. The engine should treat term extension past 72 months as a risk factor.
- **Why people don't refinance.** LendingTree found 40% of non-refinancers are satisfied with their current loan, 26% cite time and effort, and 23% expect small savings. Nearly a third of auto borrowers don't know their own interest rate [4]. Pre-filling loan data from the bureau tradeline and the servicer removes the biggest friction.
- **Where savings happen.** In Q2 2026, refinancers saved most at credit unions ($102 a month) [1], and 57% of consumers trust credit unions for refinancing, versus 33% for banks [4].

8 The reference algorithm

The design is a staged decision. Hard eligibility checks come first, then approval likelihood, then rate estimate, then economics, then a verdict with reason codes. Each stage is explainable on its own. That matters for user trust and for adverse-action style reason codes if the product ever makes or arranges credit decisions (section 12).

S0 **Gather and normalize**

Bureau tradeline (balance, APR when reported, open date, payment history), score with model and version, income, housing and other debts, VIN decode, mileage, state, servicer payoff quote, product add-ons (GAP, service contract).

S1 **Knockouts**

Branded title, commercial use, balance under $5,000, under 12 months remaining, vehicle older than 10 years or over 150,000 miles, 2+ lates of 30 days in 12 months, active bankruptcy, LTV over 130%.

S2 **Approval likelihood**

High, Medium or Low from tier, DTI, PTI, LTV, recent lates and mileage. Start with rules; replace with a calibrated model once outcome data exists.

S3 **Rate estimate**

Tier base rate plus adjustments for LTV, vehicle age, term and lender channel. Replace with live lender rate sheets or marketplace offers.

S4 **Economics**

Term-matched new payment, nominal and NPV savings net of costs and refunds, breakeven, and an optional term-extension scenario shown with total cost.

S5 **Verdict and timing**

Well positioned, Possible with conditions, Not yet (with the date to check again), Keep current loan, or Not eligible. Each comes with ranked reason codes.

S2: approval likelihood rules (starting point)

Level	All of these must hold
High	Score 661+ · DTI ≤ 45% · PTI ≤ 15% · LTV ≤ 120% · no 30-day lates in 12 months · mileage ≤ 100,000

Level	All of these must hold
`Medium`	Score 601+ · DTI ≤ 50% · PTI ≤ 20% · LTV ≤ 130% · at most one 30-day late in 12 months
`Low`	Anything else that passed S1

S3: rate estimate

```
APR_est = Base_used(tier) + adj_LTV + adj_age + adj_term + adj_channel
```

Adjustment	Bands and starting values (points)
Base by tier	Experian Q2 2026 used: 6.29 / 8.81 / 13.93 / 19.10 / 21.62 [2]
LTV	≤ 80%: −0.25 · 80–100%: 0 · 100–115%: +0.50 · 115–125%: +1.00 · over 125%: +2.00
Vehicle age	≤ 3 yrs: −0.25 · 4–6: 0 · 7–8: +0.50 · 9–10: +1.00
New term	≤ 48 mo: −0.25 · 49–72: 0 · 73–84: +1.50 (from Navy Federal's 61–72 vs. 73–84 spread [17])
Channel	Credit union: −0.50 · bank: 0 · finance company: +0.50 (direction from Experian lender-type savings [1])

The base rates are sourced. The adjustment sizes are informed starting values, not published constants. Fit them to marketplace offer data before relying on them.

S5: verdict rules

Verdict	Conditions
`Not eligible`	Any S1 knockout.
`Keep current loan`	Rate gap under 0.5 points, or term-matched NPV savings under $250. This covers most subsidized TFS loans.
`Well positioned`	Approval High · rate gap ≥ 1.0 point · NPV savings ≥ max($500, 2% of payoff) · breakeven ≤ 6 months.

Verdict	Conditions
`Possible, with conditions`	Approval Medium or better and NPV savings ≥ $250. Name the condition, for example "pay down $1,800 to reach 110% LTV."
`Not yet`	Savings exist but approval is Low. Give a check-again date: the earlier of the LTV crossover month (balance curve vs. value curve) and the seasoning month from section 3.

The thresholds ($250, $500, 2%, 6 months, 0.5 and 1.0 points) are product policy choices informed by the cost ranges above. They are not industry standards. Make them configuration values and revisit them once outcome data exists.

Reason codes

Return the top factors in plain language, ranked by impact. For example: "Your current rate (11.9%) is about 3.1 points above typical offers for your credit tier." · "You owe 118% of the car's estimated trade-in value." · "A payment 26 days late in March 2026 limits lender options until March 2027." These map naturally to the specific-reason requirement of Regulation B if the product moves into arranging credit.

9 Worked examples

Both vehicles and loans are illustrative. Values use the depreciation model in section 4 with k = 0.85.

A. 2025 Tacoma, dealer-arranged near-prime loan, 14 months in

Original loan	$42,000 at 11.90% for 72 months · payment $818.93
Today	14 payments made · payoff about $35,988 · 58 months remaining
Credit	628 at purchase (near prime) → 701 now (prime) · no lates
Value	MSRP $46,500 · δ = 0.044 · retail about $44,154 · trade about $37,531
LTV	95.9%
Refinance at 8.81% for 58 mo	$764.18/mo · saves $54.75/mo · $3,000 nominal, $2,641 NPV at 5% after $175 fees · breakeven 3.2 mo

Same at a credit union, 7.80%	$746.77/mo · saves $72.16/mo · $4,010 nominal, $3,536 NPV · breakeven 2.4 mo
If stretched to 72 months at 8.81%	$645.32/mo, but total paid $46,638 vs. $47,498 remaining on the old loan. Most of the gain disappears.
Verdict	`Well positioned`

B. 2023 RAV4 Hybrid, 36 months into a 72-month loan

Original loan	$33,000 at 9.49% for 72 months · payment $602.90
Today	Payoff about $18,824 · 36 months remaining · LTV 75%
Refinance at 7.29% for 36 mo	Saves $19.17/mo · $540 nominal, $490 NPV after $150 fees · breakeven 7.8 mo
Verdict	`Possible, with conditions` Strong equity and approval, but savings are modest because only 28% of the interest remains. The borrower should get quotes only if a lender has no fees.

Same borrower quality, very different outcomes. The difference is timing and rate gap. That is why the engine has to compute economics rather than score "credit improved" alone.

10 Reference calculator

This runs the S1–S5 logic above in the browser with the starting values. It is for checking the specification, not for consumer decisions. It opens with example A.

Static snapshot of the interactive calculator, shown with example A inputs. Use the online version to change values.

CURRENT LOAN

Original amount ($)	Current APR (%)	Original term (months)	Payments made
42000	11.9	72	14

BORROWER

Credit score (VantageScore 4.0)	30-day lates, last 12 mo	Gross monthly income ($)	Housing + other monthly debt ($)
701	0	7500	2100

VEHICLE

Model	MSRP ($)	Age (years)	Odometer (miles)
Tacoma	46500	1.2	15000

REFINANCE ASSUMPTIONS

Lender type	Upfront costs ($)	Refunds and bonuses ($)	Discount rate (%)
Bank	175	0	5

Well positioned Strong approval odds and meaningful savings after costs.

Payoff (est.)	Trade value (est.)	LTV
$36,164	$37,629	97%

Approval	Est. new APR	Monthly saving
High	8.56%	$55

NPV saving	Breakeven	PTI / DTI
$2,672	3.2 mo	10% / 38%

- Your rate (11.90%) is about 3.3 points above the estimate for your prime tier.

Term-matched comparison. Mileage slope and trade factor are placeholders; see section 4.

11 Data sources for production

Need	Source	Notes
Credit score, tradelines, payment history	Experian, Equifax or TransUnion via a reseller	Soft pull with the consumer's written authorization. Record score model and version.

Need	Source	Notes
VIN decode	NHTSA vPIC API (free, no key)	`vpic.nhtsa.dot.gov/api` returns year, make, model, trim, engine [38]
Recalls	NHTSA Recalls API	`api.nhtsa.gov/recalls/recallsByVehicle?make=&model=&modelYear=`; check VIN-level status on NHTSA or Toyota's lookup [38]
Vehicle value	J.D. Power Valuation Services, Black Book, KBB (licensed)	Pull clean trade and clean retail. Keep both.
Market rates	Lender rate sheets, marketplace partners; Experian quarterly report for priors	Refresh at least monthly
Payoff	Servicer payoff quote (TFS account, app or phone)	Quotes are date-specific because interest accrues daily [26]
State fees and taxes	DMV fee schedules	Maintain a versioned table by state
Wholesale trend	Cox Automotive Manheim index	Monthly; use segment moves to stress LTV [7]

12 Regulatory and compliance considerations

This is a summary of research, not legal advice. The product's obligations depend on whether it only informs consumers, refers them to lenders, or takes part in credit decisions. Engage counsel before launch.

- **FCRA.** Pulling a consumer's report for this purpose requires a permissible purpose, typically the consumer's written instruction. Unsolicited marketing based on bureau data requires a prescreened firm offer of credit with Regulation V opt-out notices. Regulators have criticized trigger-lead practices that did not make firm offers [39].
- **ECOA and Regulation B.** Creditors must give specific principal reasons for adverse action. The CFPB's 2022 circular said complex models do not excuse this [40]. That circular was among guidance the CFPB withdrew in May 2025 [41], but the underlying Regulation B requirement still

stands. "Creditor" can include parties that regularly take part in credit decisions, so a marketplace that pre-qualifies on lenders' behalf should assume these rules apply.

- **Disparate impact.** The CFPB finalized a rule in April 2026, effective July 21, 2026, stating that ECOA does not support disparate-impact liability [42]. Commentators note courts may not agree, and state laws and attorneys general can still pursue disparate-impact theories. The TMCC consent order [24] shows how pricing discretion was treated before. Testing outcomes across groups remains prudent for bank partners and state exposure.
- **Model risk.** In April 2026 the Fed, OCC and FDIC replaced SR 11-7 with SR 26-2, which keeps independent validation and governance and scales expectations to materiality [43]. Bank partners will expect documentation that meets it.
- **TILA §1615.** Unearned interest must be refunded on prepayment, and the Rule of 78s is barred on precomputed loans over 61 months [27]. The payoff logic should handle precomputed contracts correctly.
- **Other.** State broker and lead-generator licensing (for example California), the FTC Safeguards Rule for customer financial data, UDAAP standards on savings claims ("save $X" must be term-matched and net of fees), and trademark rules on using the Toyota name.

13 Validation plan

1. **Unit-test the finance math** against known amortization tables, precomputed contracts and per-diem payoffs. These must match to the cent.
2. **Backtest the rate estimate** against real offers. Track mean absolute error in APR points by tier, overall and for Toyota models. Target under 0.75 points at prime.
3. **Calibrate approval likelihood.** Once you have approve/decline outcomes, fit a logistic or gradient-boosted model. Report AUC and calibration (Brier score, reliability curve). Keep the rules as a challenger.
4. **Check the value model** against guidebook values by VIN. Fit β and k from data and test LTV error at the 100% and 125% thresholds, where errors change outcomes.
5. **Track realized outcomes.** Did "Well positioned" users get funded at or below the estimate? Did "Not yet" dates prove accurate?
6. **Monitor drift** quarterly against Experian tier rates, the Manheim index and NY Fed delinquency. Refresh the Toyota depreciation table annually when iSeeCars and ALG publish.
7. **Fair lending review.** Compare verdicts and estimated APRs across demographic groups (using proxies where permitted) and document less discriminatory alternatives.

14 Open questions and data gaps

- No lender publishes which valuation book and condition level it uses for refinance LTV. Collect this from partners directly.
- No peer-reviewed study sets an auto-specific PTI or DTI default threshold. The values here are industry practice and ABS benchmarks.
- The rate adjustments and verdict thresholds are informed starting values. They need fitting on offer and outcome data.
- The CFPB has not published a study of auto refinance savings. The best evidence is Experian aggregates, the LendingTree study and the Lautier et al. paper.
- 2026 depreciation figures were not found for the Camry, Crown, Sequoia, Grand Highlander or bZ.
- Whether the 2016 TMCC markup cap is still in force was not confirmed. Consent orders often expire.
- Edmunds' Q3 2026 negative-equity figures were not yet available. Q2 2026 is used.

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